

Novelty Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Novelty, Transience, and Resonance Analysis

PENPAL Project

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1 Overview

This notebook compares **novelty metrics** across all three experimental conditions:

- **Human-AI:** Human co-writing with an LLM
- **Human-Human:** Two humans co-writing together
- **AI-AI:** Two AI agents co-writing (simulated baseline)

Key Metrics (Information-Theoretic Framework):

- **Novelty (Surprisal):** How unexpected is a turn given past context? Higher = more surprising/novel
- **Transience:** How quickly does a turn's influence fade? Higher = less lasting impact

- **Resonance:** Novelty minus Transience — net forward influence on the narrative

Key Questions: 1. Do humans produce higher novelty than AI? 2. Does novelty lead to greater resonance in human contributions? 3. How do Human-Human dynamics compare to Human-AI? 4. Is the AI-AI baseline more predictable (lower novelty)?

2 Load Data

```
## Loaded human-ai from novelty_scores.csv
## Loaded human-human from novelty_scores.csv
## Loaded ai-ai from novelty_scores.csv

##
## Loaded conditions: human-ai, human-human, ai-ai
```

3 Schema Normalization

```
##
## Analysis dataset:

## # A tibble: 3 x 3
##   condition  n_stories n_turns
##   <fct>      <int>   <int>
## 1 Human-AI      88     704
## 2 Human-Human   38     304
## 3 AI-AI        40     320

##
## Available columns:

## [1] "timestamp"           "author_1"
## [3] "author_2"           "conversation_id"
## [5] "respondent_id"       "interaction_count"
## [7] "llm_type"            "turn"
## [9] "starter"             "author_1_surprise"
## [11] "author_2_surprise"   "user_surprise_raw"
## [13] "ai_surprise_raw"     "user_entropy"
## [15] "ai_entropy"          "author_1_transience"
## [17] "author_2_transience" "user_transience_raw"
## [19] "ai_transience_raw"  "condition"
## [21] "respondent_id_u1"    "respondent_id_u2"
## [23] "author_1_surprise_raw" "author_2_surprise_raw"
## [25] "author_1_entropy"    "author_2_entropy"
## [27] "author_1_transience_raw" "author_2_transience_raw"
## [29] "model_id"            "author_1_resonance"
## [31] "author_2_resonance"
```

3.1 Build Long-Format Dataset with Author Factor

Following the within-condition approach, we stack Author 1 and Author 2 metrics into a long format with `author_role` as a factor. This enables proper testing of asymmetry (whether metrics differ by author role) within and across conditions.

```
## === Long-Format Dataset ===

## # A tibble: 6 x 7
```

```
## condition author_role n_obs n_stories mean_surprise mean_transience
## <fct> <fct> <int> <int> <dbl> <dbl>
## 1 Human-AI author_1 704 88 -1.42 -1.03
## 2 Human-AI author_2 704 88 -2.06 -0.671
## 3 Human-Human author_1 304 38 -1.68 -0.935
## 4 Human-Human author_2 304 38 -1.73 -0.911
## 5 AI-AI author_1 320 40 -2.28 -1.45
## 6 AI-AI author_2 320 40 -2.37 -1.40
## # i 1 more variable: mean_resonance <dbl>
```

3.2 Standardize Within Conversation $\times$ Author

Z-score metrics within each (conversation_id $\times$ author_role) cell to ensure comparable coefficients across author roles and conditions.

```
## Standardized dataset: 2656 observations
```

4 Descriptive Statistics

Table 1: Novelty metrics by condition and author role

condition	author_role	n_stories	n_obs	surprise_mean	surprise_sd	transience_mean	transience_sd	resonance_mean	resonance_sd
Human-AI	author_1	88	704	-1.421	0.714	-1.033	0.441	-0.389	0.777
Human-AI	author_2	88	704	-2.061	0.680	-0.671	0.460	-1.391	0.696
Human-Human	author_1	38	304	-1.683	0.674	-0.935	0.543	-0.748	0.788
Human-Human	author_2	38	304	-1.728	0.639	-0.911	0.591	-0.817	0.740
AI-AI	author_1	40	320	-2.280	0.609	-1.448	0.595	-0.832	0.831
AI-AI	author_2	40	320	-2.366	0.697	-1.397	0.523	-0.969	0.866

5 Within-Condition Asymmetry Analysis

For each condition, we test whether novelty metrics differ by author role using **metric** ~ **author_role** models. Significant effects indicate asymmetry (e.g., Author 1 shows higher novelty than Author 2).

5.1 Novelty (Surprisal) Asymmetry

```
##
## === Human-AI: Novelty Asymmetry ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: surprise ~ author_role + (1 | conversation_id)
## Data: df_cond
##
## REML criterion at convergence: 2939.5
##
## Scaled residuals:
## Min 1Q Median 3Q Max
## -3.0546 -0.6113 0.0343 0.6314 4.7378
```

```

##
## Random effects:
##   Groups      Name      Variance Std.Dev.
## conversation_id (Intercept) 0.04491 0.2119
## Residual              0.44165 0.6646
## Number of obs: 1408, groups: conversation_id, 88
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  -1.74135    0.02871   87.00000  -60.66  <2e-16 ***
## author_role1    0.31995    0.01771 1319.00000   18.07  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## author_role1 0.000
##
## --- Author Means ---
##   author_role emmean      SE  df lower.CL upper.CL
##   author_1    -1.42 0.0337 164    -1.49    -1.35
##   author_2    -2.06 0.0337 164    -2.13    -1.99
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## --- Asymmetry Test ---
##   contrast      estimate      SE  df t.ratio p.value
##   author_1 - author_2    0.64 0.0354 1319  18.065  <.0001
##
## Degrees-of-freedom method: kenward-roger
##
## === Human-Human: Novelty Asymmetry ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: surprise ~ author_role + (1 | conversation_id)
##   Data: df_cond
##
## REML criterion at convergence: 1175.5
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.0661 -0.5756  0.0635  0.6174  3.3378
##
## Random effects:
##   Groups      Name      Variance Std.Dev.
## conversation_id (Intercept) 0.06502 0.2550
## Residual              0.36738 0.6061
## Number of obs: 608, groups: conversation_id, 38
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  -1.70578    0.04812   37.00000  -35.451  <2e-16 ***
## author_role1    0.02231    0.02458  569.00000   0.908    0.364

```

```

## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr)
## author_rol1 0.000
##
## --- Author Means ---
##  author_role emmean      SE   df lower.CL upper.CL
##  author_1    -1.68 0.054 58.6   -1.79   -1.58
##  author_2    -1.73 0.054 58.6   -1.84   -1.62
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## --- Asymmetry Test ---
##  contrast      estimate      SE   df t.ratio p.value
##  author_1 - author_2  0.0446 0.0492 569   0.908  0.3644
##
## Degrees-of-freedom method: kenward-roger
##
## === AI-AI: Novelty Asymmetry ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: surprise ~ author_role + (1 | conversation_id)
## Data: df_cond
##
## REML criterion at convergence: 1195.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.4690 -0.6376  0.0130  0.6685  3.0979
##
## Random effects:
##  Groups      Name      Variance Std.Dev.
##  conversation_id (Intercept) 0.09382  0.3063
##  Residual              0.33680  0.5803
## Number of obs: 640, groups:  conversation_id, 40
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  -2.32284    0.05359 39.00000 -43.346  <2e-16 ***
## author_role1   0.04315    0.02294 599.00000   1.881   0.0604 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr)
## author_rol1 0.000
##
## --- Author Means ---
##  author_role emmean      SE   df lower.CL upper.CL
##  author_1    -2.28 0.0583 54.5   -2.40   -2.16
##  author_2    -2.37 0.0583 54.5   -2.48   -2.25

```

```
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## --- Asymmetry Test ---
## contrast          estimate      SE  df t.ratio p.value
## author_1 - author_2  0.0863 0.0459 599   1.881  0.0604
##
## Degrees-of-freedom method: kenward-roger
```

Table 2: Novelty asymmetry by condition (Author 1 - Author 2)

condition	metric	author_1_mean	author_2_mean	difference	se	t	p	sig	interpretation
Human-AI	Novelty	-1.421	-2.061	0.640	0.035	18.065	0.000	***	Author 1 > Author 2
Human-Human	Novelty	-1.683	-1.728	0.045	0.049	0.908	0.364		No asymmetry
AI-AI	Novelty	-2.280	-2.366	0.086	0.046	1.881	0.060	.	No asymmetry

5.2 Transience Asymmetry

```
##
## === Human-AI: Transience Asymmetry ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: transience ~ author_role + (1 | conversation_id)
## Data: df_cond
##
## REML criterion at convergence: 1669.2
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.9185 -0.6469  0.0220  0.6733  3.0196
##
## Random effects:
## Groups          Name          Variance Std.Dev.
## conversation_id (Intercept) 0.02761  0.1662
## Residual              0.17574  0.4192
## Number of obs: 1408, groups: conversation_id, 88
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  -0.85159    0.02094   87.00000  -40.66  <2e-16 ***
## author_role1  -0.18109    0.01117  1319.00000  -16.21  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## author_role1 0.000
##
## === Human-Human: Transience Asymmetry ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
```

```

## lmerModLmerTest]
## Formula: transience ~ author_role + (1 | conversation_id)
## Data: df_cond
##
## REML criterion at convergence: 996.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -6.7284 -0.5153  0.0548  0.6314  3.5100
##
## Random effects:
## Groups           Name          Variance Std.Dev.
## conversation_id (Intercept) 0.05066  0.2251
## Residual              0.27294  0.5224
## Number of obs: 608, groups: conversation_id, 38
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  -0.92312    0.04221  37.00000 -21.867  <2e-16 ***
## author_role1 -0.01233    0.02119 569.00000  -0.582    0.561
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## author_role1 0.000
##
## === AI-AI: Transience Asymmetry ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: transience ~ author_role + (1 | conversation_id)
## Data: df_cond
##
## REML criterion at convergence: 1081
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -6.0926 -0.5953  0.0313  0.6307  2.7895
##
## Random effects:
## Groups           Name          Variance Std.Dev.
## conversation_id (Intercept) 0.008016 0.08953
## Residual              0.305683 0.55289
## Number of obs: 640, groups: conversation_id, 40
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  -1.42240    0.02604  39.00000 -54.63  <2e-16 ***
## author_role1 -0.02513    0.02185 598.99999  -1.15    0.251
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)

```

```
## author_rol1 0.000
```

Table 3: Transience asymmetry by condition (Author 1 - Author 2)

condition	metric	author_1_mean	author_2_mean	difference	se	t	p	sig	interpretation
Human-AI	Transience	-1.033	-0.671	-0.362	0.022	-16.209	0.000	***	Author 2 > Author 1
Human-Human	Transience	-0.935	-0.911	-0.025	0.042	-0.582	0.561		No asymmetry
AI-AI	Transience	-1.448	-1.397	-0.050	0.044	-1.150	0.251		No asymmetry

5.3 Resonance Asymmetry

```
##
## === Human-AI: Resonance Asymmetry ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: resonance ~ author_role + (1 | conversation_id)
## Data: df_cond
##
## REML criterion at convergence: 3145.1
##
## Scaled residuals:
## Min 1Q Median 3Q Max
## -3.3110 -0.6357 0.0253 0.6667 3.3883
##
## Random effects:
## Groups Name Variance Std.Dev.
## conversation_id (Intercept) 0.01228 0.1108
## Residual 0.53223 0.7295
## Number of obs: 1408, groups: conversation_id, 88
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) -0.88976 0.02275 87.00000 -39.11 <2e-16 ***
## author_role1 0.50104 0.01944 1319.00000 25.77 <2e-16 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
## (Intr)
## author_rol1 0.000
##
## === Human-Human: Resonance Asymmetry ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: resonance ~ author_role + (1 | conversation_id)
## Data: df_cond
##
## REML criterion at convergence: 1407.1
##
## Scaled residuals:
```



```

##      Min      1Q  Median      3Q      Max
## -4.4611 -0.5852  0.0088  0.6461  4.1471
##
## Random effects:
##   Groups      Name      Variance Std.Dev.
## conversation_id (Intercept) 0.0000  0.0000
## Residual                0.5844  0.7645
## Number of obs: 608, groups: conversation_id, 38
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  -0.78266    0.03100 606.00000 -25.244  <2e-16 ***
## author_role1  0.03464    0.03100 606.00000  1.117   0.264
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## author_rol1 0.000
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
##
##
## === AI-AI: Resonance Asymmetry ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: resonance ~ author_role + (1 | conversation_id)
## Data: df_cond
##
## REML criterion at convergence: 1602.5
##
## Scaled residuals:
##      Min      1Q  Median      3Q      Max
## -4.6313 -0.6674  0.0404  0.6305  4.2901
##
## Random effects:
##   Groups      Name      Variance Std.Dev.
## conversation_id (Intercept) 0.04309  0.2076
## Residual                0.67754  0.8231
## Number of obs: 640, groups: conversation_id, 40
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  -0.90044    0.04622  39.00000 -19.484  <2e-16 ***
## author_role1  0.06829    0.03254 599.00000  2.099   0.0363 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## author_rol1 0.000

```

Table 4: Resonance asymmetry by condition (Author 1 - Author 2)

condition	metric	author_1_mean	author_2_mean	difference	se	t	p	sig	interpretation
Human-AI	Resonance	-0.389	-1.391	1.002	0.039	25.771	0.000	***	Author 1 > Author 2
Human-Human	Resonance	-0.748	-0.817	0.069	0.062	1.117	0.264		No asymmetry
AI-AI	Resonance	-0.832	-0.969	0.137	0.065	2.099	0.036	*	Author 1 > Author 2

5.4 Summary of Within-Condition Asymmetry

Table 5: Summary: Asymmetry in novelty metrics by condition

Condition	Metric	Author 1	Author 2	Diff	SE	p	
Human-AI	Novelty	-1.421	-2.061	0.640	0.035	0.000	***
Human-Human	Novelty	-1.683	-1.728	0.045	0.049	0.364	
AI-AI	Novelty	-2.280	-2.366	0.086	0.046	0.060	.
Human-AI	Transience	-1.033	-0.671	-0.362	0.022	0.000	***
Human-Human	Transience	-0.935	-0.911	-0.025	0.042	0.561	
AI-AI	Transience	-1.448	-1.397	-0.050	0.044	0.251	
Human-AI	Resonance	-0.389	-1.391	1.002	0.039	0.000	***
Human-Human	Resonance	-0.748	-0.817	0.069	0.062	0.264	
AI-AI	Resonance	-0.832	-0.969	0.137	0.065	0.036	*

6 Cross-Condition Comparison

6.1 Test if Asymmetry Differs Across Conditions

Using **metric** ~ **author_role** × **condition** interactions to test whether the pattern of asymmetry (Author 1 vs Author 2 difference) varies across conditions.

```
## === Novelty: Author × Condition Interaction ===

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: surprise ~ author_role * condition + (1 | conversation_id)
## Data: df_std
##
## REML criterion at convergence: 5334.8
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.1710 -0.6046  0.0397  0.6423  4.9505
##
## Random effects:
## Groups          Name          Variance Std.Dev.
## conversation_id (Intercept) 0.0612   0.2474
## Residual                0.3994   0.6320
## Number of obs: 2656, groups: conversation_id, 166
##
```

```

## Fixed effects:
##
##               Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      -1.92332    0.02450 162.99999 -78.516 < 2e-16 ***
## author_role1       0.12847    0.01319 2487.00001   9.744 < 2e-16 ***
## condition1         0.18198    0.03044 162.99999   5.979 1.38e-08 ***
## condition2         0.21754    0.03682 162.99999   5.908 1.96e-08 ***
## author_role1:condition1  0.19148    0.01638 2487.00001  11.687 < 2e-16 ***
## author_role1:condition2 -0.10616    0.01982 2487.00001  -5.356 9.28e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) athr_1 cndtn1 cndtn2 at_1:1
## author_rol1  0.000
## condition1  -0.367  0.000
## condition2   0.173  0.000 -0.430
## athr_rl1:c1  0.000 -0.367  0.000  0.000
## athr_rl1:c2  0.000  0.173  0.000  0.000 -0.430
##
## === Novelty Means by Condition × Author ===
##   condition  author_role emmean      SE df lower.CL upper.CL
## Human-AI    author_1     -1.42 0.0355 270    -1.49    -1.35
## Human-Human author_1     -1.68 0.0541 270    -1.79    -1.58
## AI-AI        author_1     -2.28 0.0527 270    -2.38    -2.18
## Human-AI    author_2     -2.06 0.0355 270    -2.13    -1.99
## Human-Human author_2     -1.73 0.0541 270    -1.83    -1.62
## AI-AI        author_2     -2.37 0.0527 270    -2.47    -2.26
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## === Asymmetry Within Each Condition ===
## condition = Human-AI:
##   contrast      estimate      SE df t.ratio p.value
## author_1 - author_2  0.6399 0.0337 2487  18.997 <.0001
##
## condition = Human-Human:
##   contrast      estimate      SE df t.ratio p.value
## author_1 - author_2  0.0446 0.0513 2487   0.871  0.3841
##
## condition = AI-AI:
##   contrast      estimate      SE df t.ratio p.value
## author_1 - author_2  0.0863 0.0500 2487   1.727  0.0842
##
## Degrees-of-freedom method: kenward-roger
##
## === Does Asymmetry Differ Across Conditions? ===
##   condition_pairwise  author_role_pairwise estimate      SE df t.ratio
## (Human-AI) - (Human-Human) author_1 - author_2   0.5953 0.0613 2487   9.705
## (Human-AI) - (AI-AI)      author_1 - author_2   0.5536 0.0603 2487   9.187
## (Human-Human) - (AI-AI)   author_1 - author_2  -0.0417 0.0716 2487  -0.582

```

```

## p.value
## <.0001
## <.0001
## 0.5604
##
## Degrees-of-freedom method: kenward-roger

## === Transience: Author × Condition Interaction ===

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: transience ~ author_role * condition + (1 | conversation_id)
## Data: df_std
##
## REML criterion at convergence: 3839.1
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -7.4053 -0.5862  0.0321  0.6388  3.8863
##
## Random effects:
##      Groups             Name             Variance Std.Dev.
## conversation_id (Intercept) 0.02813  0.1677
## Residual                    0.22927  0.4788
## Number of obs: 2656, groups: conversation_id, 166
##
## Fixed effects:
##
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      -1.06570    0.01720 163.00000 -61.972 < 2e-16 ***
## author_role1      -0.07285    0.00999 2487.00000  -7.292 4.08e-13 ***
## condition1         0.21411    0.02137 163.00000  10.021 < 2e-16 ***
## condition2         0.14258    0.02585 163.00000   5.516 1.34e-07 ***
## author_role1:condition1 -0.10824    0.01241 2487.00000  -8.720 < 2e-16 ***
## author_role1:condition2  0.06052    0.01502 2487.00000   4.030 5.74e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) athr_1 cndtn1 cndtn2 at_1:1
## author_rol1  0.000
## condition1  -0.367  0.000
## condition2   0.173  0.000 -0.430
## athr_rl1:c1  0.000 -0.367  0.000  0.000
## athr_rl1:c2  0.000  0.173  0.000  0.000 -0.430
##
## === Transience Asymmetry by Condition ===

## condition = Human-AI:
##      contrast          estimate      SE    df t.ratio p.value
## author_1 - author_2 -0.3622 0.0255 2487 -14.191 <.0001
##
## condition = Human-Human:
##      contrast          estimate      SE    df t.ratio p.value
## author_1 - author_2 -0.0247 0.0388 2487  -0.635  0.5256
##

```

```

## condition = AI-AI:
## contrast          estimate      SE    df t.ratio p.value
## author_1 - author_2 -0.0503 0.0379 2487  -1.328  0.1844
##
## Degrees-of-freedom method: kenward-roger

## === Resonance: Author × Condition Interaction ===

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: resonance ~ author_role * condition + (1 | conversation_id)
## Data: df_std
##
## REML criterion at convergence: 6183.6
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -5.1324 -0.6213  0.0324  0.6475  4.1956
##
## Random effects:
## Groups          Name          Variance Std.Dev.
## conversation_id (Intercept) 0.01395  0.1181
## Residual                0.58202  0.7629
## Number of obs: 2656, groups: conversation_id, 166
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    -0.85762    0.01872 163.00000 -45.812 < 2e-16 ***
## author_role1      0.20132    0.01592 2487.00000  12.649 < 2e-16 ***
## condition1     -0.03214    0.02326 163.00000  -1.382  0.1690
## condition2      0.07496    0.02814 163.00000   2.664  0.0085 **
## author_role1:condition1  0.29972    0.01978 2487.00000  15.155 < 2e-16 ***
## author_role1:condition2 -0.16668    0.02393 2487.00000  -6.967 4.14e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) athr_1 cndtn1 cndtn2 at_1:1
## author_rol1  0.000
## condition1 -0.367  0.000
## condition2  0.173  0.000 -0.430
## athr_rl1:c1  0.000 -0.367  0.000  0.000
## athr_rl1:c2  0.000  0.173  0.000  0.000 -0.430
##
## === Resonance Asymmetry by Condition ===

## condition = Human-AI:
## contrast          estimate      SE    df t.ratio p.value
## author_1 - author_2  1.0021 0.0407 2487  24.644 <.0001
##
## condition = Human-Human:
## contrast          estimate      SE    df t.ratio p.value
## author_1 - author_2  0.0693 0.0619 2487   1.120  0.2630
##
## condition = AI-AI:

```

```
## contrast          estimate      SE   df t.ratio p.value
## author_1 - author_2  0.1366 0.0603 2487   2.264  0.0236
##
## Degrees-of-freedom method: kenward-roger
```

7 Novelty-Resonance Relationship

7.1 Does the Innovation Bias (Novelty → Resonance Slope) Differ by Author and Condition?

```
##
## === Human-AI: Innovation Bias (Novelty → Resonance) ===
## Slopes by author:
## author_role surprise.trend      SE   df lower.CL upper.CL
## author_1          0.945 0.0227 1394    0.90   0.989
## author_2          0.817 0.0240 1400    0.77   0.865
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## Slope difference test:
## contrast          estimate      SE   df t.ratio p.value
## author_1 - author_2  0.127 0.0328 1389   3.879  0.0001
##
## Degrees-of-freedom method: kenward-roger
##
## === Human-Human: Innovation Bias (Novelty → Resonance) ===
## Slopes by author:
## author_role surprise.trend      SE   df lower.CL upper.CL
## author_1          0.927 0.0463 598    0.836   1.018
## author_2          0.847 0.0501 604    0.749   0.946
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## Slope difference test:
## contrast          estimate      SE   df t.ratio p.value
## author_1 - author_2  0.0796 0.0664 589    1.199  0.2308
##
## Degrees-of-freedom method: kenward-roger
##
## === AI-AI: Innovation Bias (Novelty → Resonance) ===
## Slopes by author:
## author_role surprise.trend      SE   df lower.CL upper.CL
## author_1          0.968 0.0520 628    0.866   1.07
## author_2          1.010 0.0461 555    0.920   1.10
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## Slope difference test:
## contrast          estimate      SE   df t.ratio p.value
## author_1 - author_2  -0.042 0.0681 628   -0.616  0.5383
```

```
##
## Degrees-of-freedom method: kenward-roger
```

Table 6: Innovation bias (Novelty→Resonance slope) asymmetry
by condition

condition	author_1_slope	author_2_slope	slope_diff	se	t	p	sig	interpretation
Human-AI	0.945	0.817	0.127	0.033	3.879	0.000	***	Author 1 slope > Author 2
Human-Human	0.927	0.847	0.080	0.066	1.199	0.231		No slope difference
AI-AI	0.968	1.010	-0.042	0.068	-0.616	0.538		No slope difference

7.2 Cross-Condition Innovation Bias Comparison

```
## === Full Model: Novelty × Author × Condition ===

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: resonance ~ surprise * author_role * condition + (1 | conversation_id)
## Data: df_std
##
## REML criterion at convergence: 3811.2
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.8290 -0.6384 -0.0328  0.5729  7.3050
##
## Random effects:
##   Groups             Name             Variance Std.Dev.
##   conversation_id (Intercept) 0.02189  0.1480
##   Residual                  0.22632  0.4757
## Number of obs: 2656, groups:  conversation_id, 166
##
## Fixed effects:
##                                     Estimate Std. Error      df t value
## (Intercept)                      0.92758    0.03601 1414.28908  25.760
## surprise                        0.92222    0.01617 2578.09283  57.043
## author_role1                     0.12807    0.03219 2574.36376   3.979
## condition1                      -0.30942    0.04268 1361.16142 -7.250
## condition2                      -0.21616    0.05076 1301.86071 -4.258
## surprise:author_role1             0.02677    0.01537 2586.03905   1.741
## surprise:condition1              -0.04479    0.01957 2610.77912 -2.289
## surprise:condition2              -0.04578    0.02433 2591.25492 -1.882
## author_role1:condition1           0.20101    0.03832 2591.96646   5.246
## author_role1:condition2          -0.04125    0.04517 2576.13035 -0.913
## surprise:author_role1:condition1  0.03570    0.01883 2603.98298   1.896
## surprise:author_role1:condition2  0.01529    0.02320 2587.03702   0.659
##                                     Pr(>|t|)
## (Intercept)                      < 2e-16 ***
## surprise                        < 2e-16 ***
## author_role1                     7.11e-05 ***
## condition1                       6.97e-13 ***
```

```

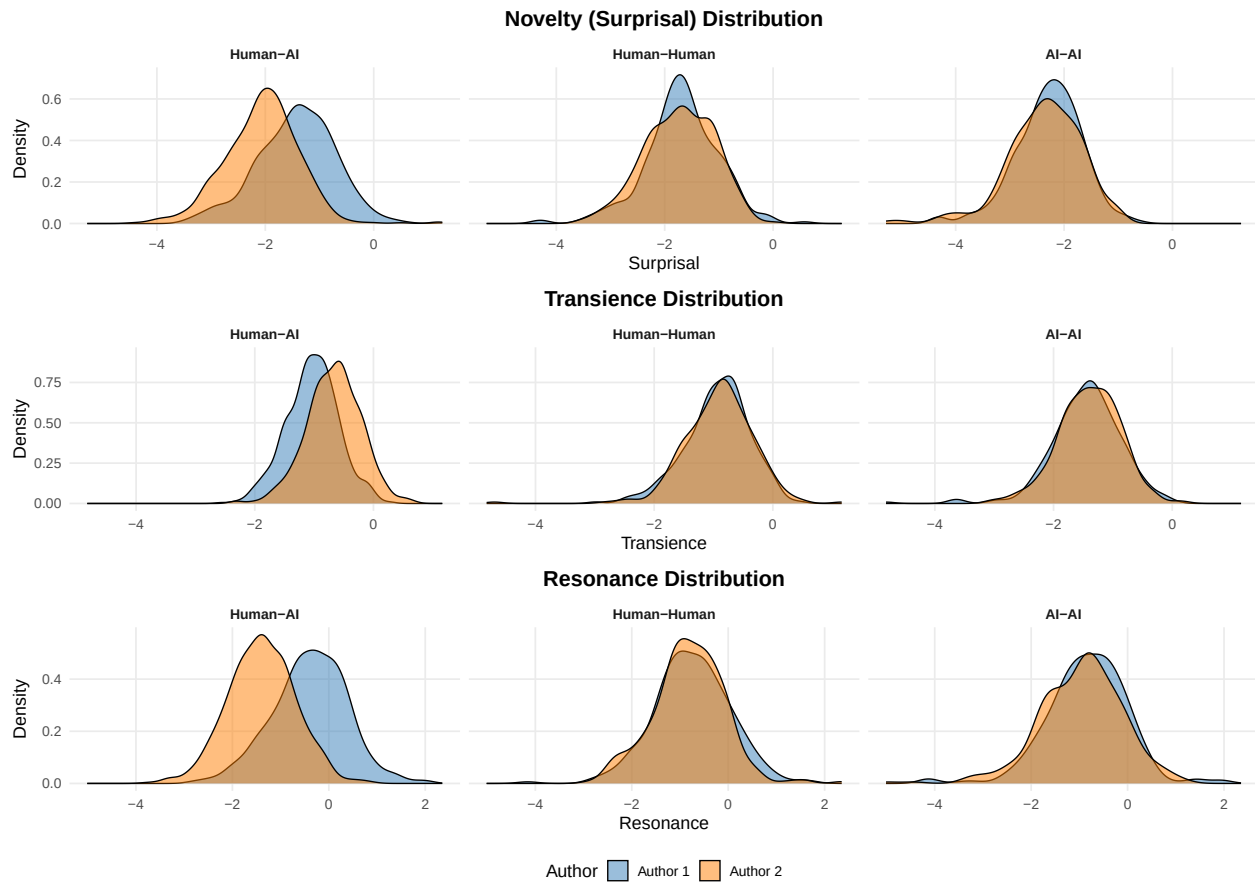
## condition2                2.21e-05 ***
## surprise:author_role1      0.0818 .
## surprise:condition1        0.0222 *
## surprise:condition2        0.0600 .
## author_role1:condition1    1.68e-07 ***
## author_role1:condition2    0.3612
## surprise:author_role1:condition1 0.0581 .
## surprise:author_role1:condition2 0.5099
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr)  surprs  athr_1  cndtn1  cndtn2  srp:_1  srpr:1  srpr:2  at_1:1
## surprise      0.886
## author_rol1 -0.038 -0.023
## condition1  -0.502 -0.438 -0.080
## condition2  -0.009  0.059 -0.030 -0.235
## srprs:thr_1 -0.020  0.003  0.939 -0.059 -0.046
## srprs:cndt1 -0.429 -0.442 -0.061  0.874 -0.264 -0.025
## srprs:cndt2  0.055  0.176 -0.046 -0.253  0.875 -0.065 -0.401
## athr_rl1:c1 -0.080 -0.062 -0.490 -0.121  0.105 -0.435 -0.083  0.092
## athr_rl1:c2 -0.030 -0.049 -0.022  0.105 -0.060  0.073  0.097 -0.054 -0.231
## srprs:t_1:1 -0.057 -0.025 -0.423 -0.075  0.090 -0.408 -0.016  0.068  0.932
## srprs:t_1:2 -0.043 -0.065  0.068  0.086 -0.049  0.183  0.069 -0.040 -0.291
##      at_1:2 s:_1:1
## surprise
## author_rol1
## condition1
## condition2
## srprs:thr_1
## srprs:cndt1
## srprs:cndt2
## athr_rl1:c1
## athr_rl1:c2
## srprs:t_1:1 -0.304
## srprs:t_1:2  0.935 -0.421
##
## === Innovation Slopes by Condition × Author ===
##      condition  author_role  surprise.trend      SE    df lower.CL upper.CL
## Human-AI      author_1      0.940 0.0261 2638    0.889    0.991
## Human-Human  author_1      0.919 0.0419 2631    0.836    1.001
## AI-AI         author_1      0.989 0.0454 2642    0.900    1.078
## Human-AI      author_2      0.815 0.0276 2644    0.761    0.869
## Human-Human  author_2      0.834 0.0452 2636    0.746    0.923
## AI-AI         author_2      1.037 0.0409 2596    0.957    1.117
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## === Slope Asymmetry Within Each Condition ===
## condition = Human-AI:
##      contrast      estimate      SE    df t.ratio p.value

```



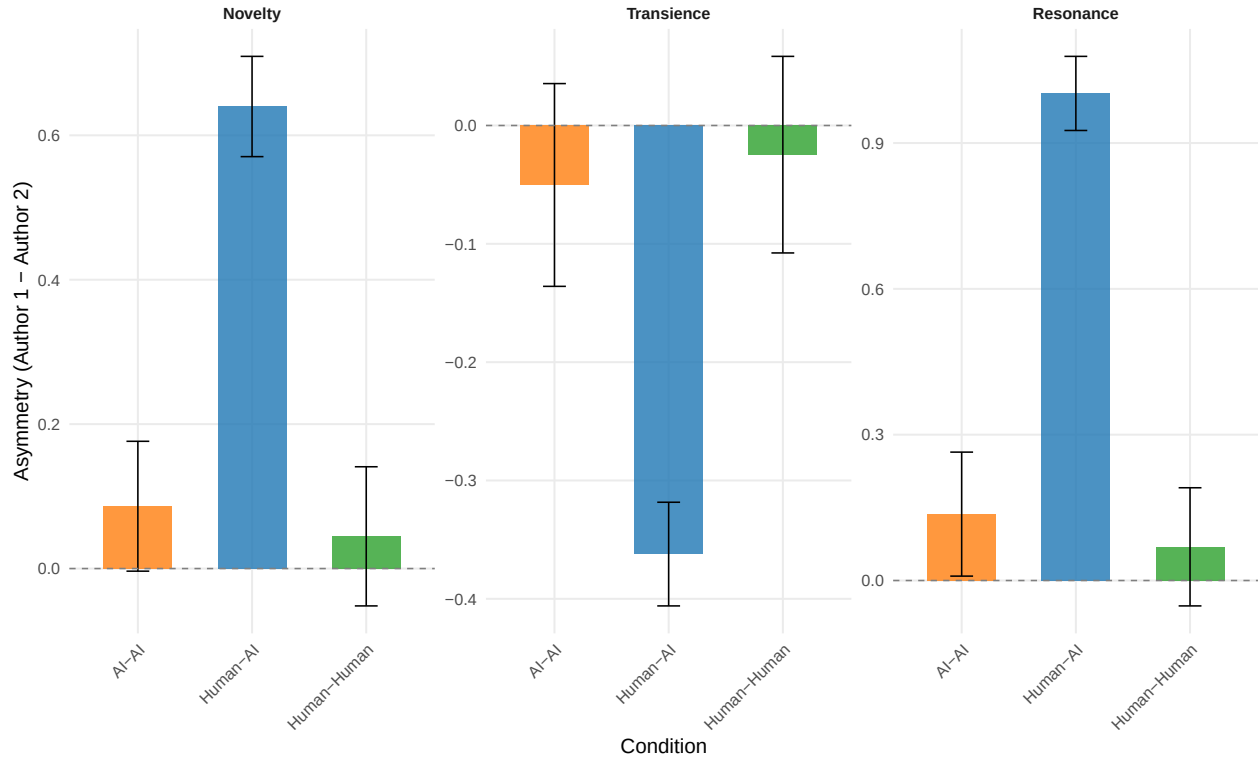
```
## author_1 - author_2    0.1249 0.0377 2631    3.314 0.0009
##
## condition = Human-Human:
## contrast      estimate      SE    df t.ratio p.value
## author_1 - author_2    0.0841 0.0602 2592    1.397 0.1625
##
## condition = AI-AI:
## contrast      estimate      SE    df t.ratio p.value
## author_1 - author_2   -0.0485 0.0589 2563   -0.823 0.4109
##
## Degrees-of-freedom method: kenward-roger
```

8 Visualization



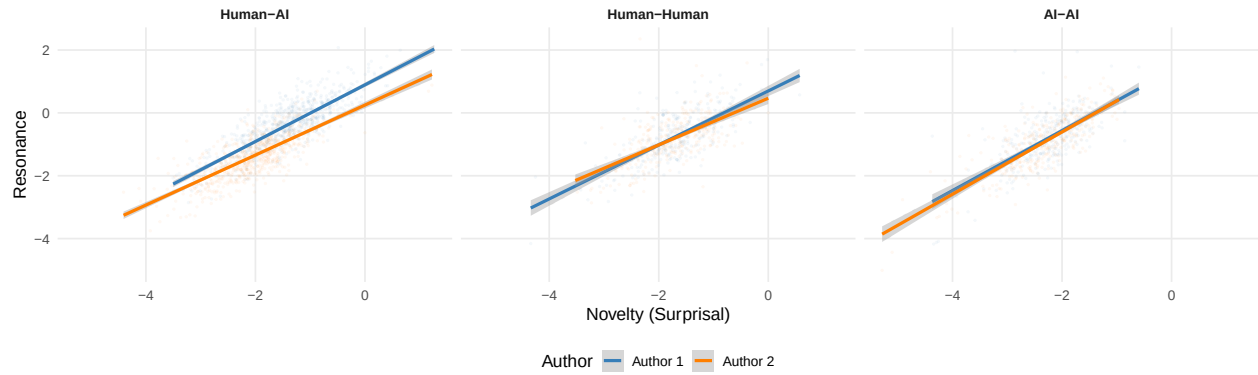
Novelty Metric Asymmetry by Condition

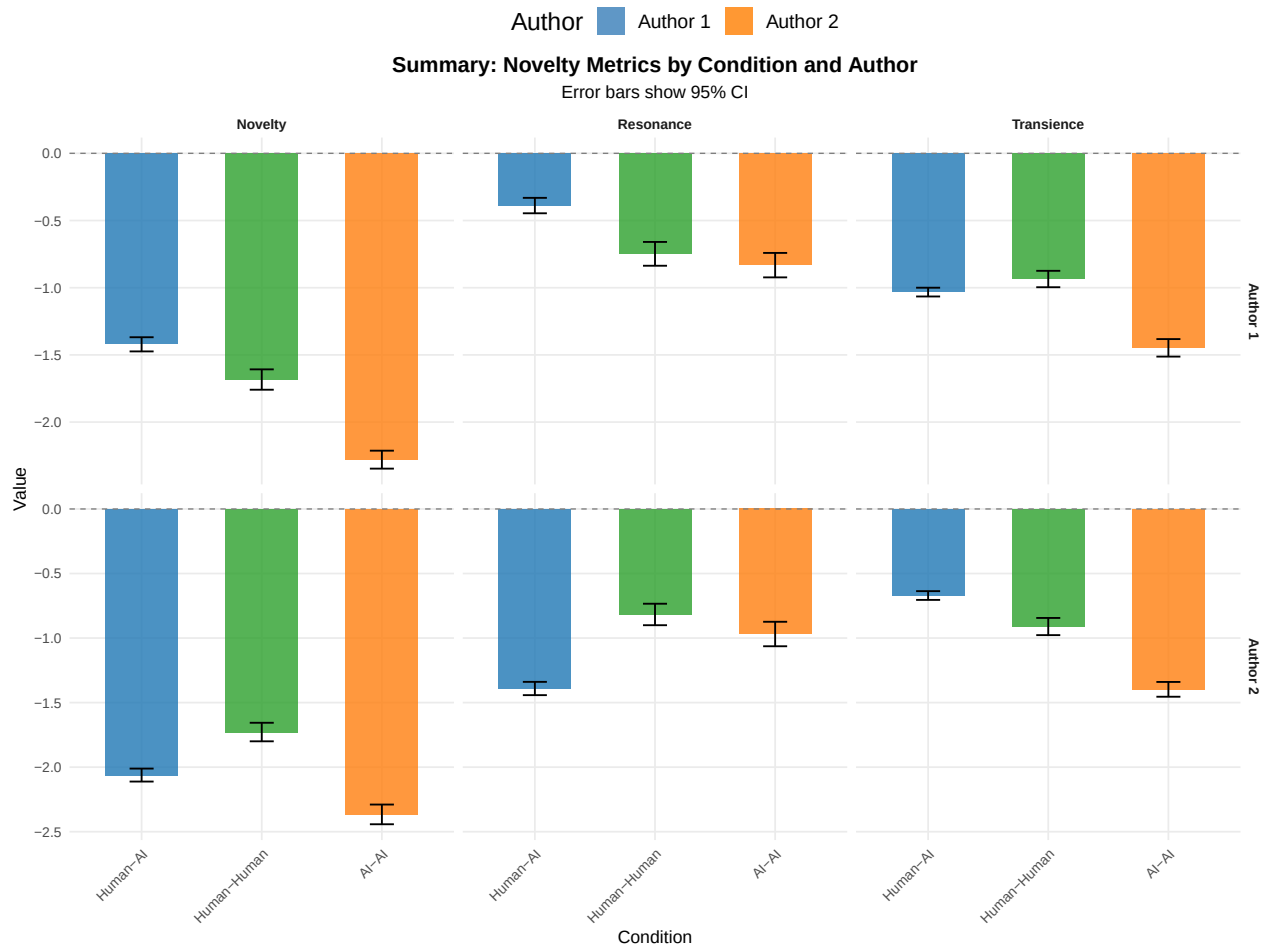
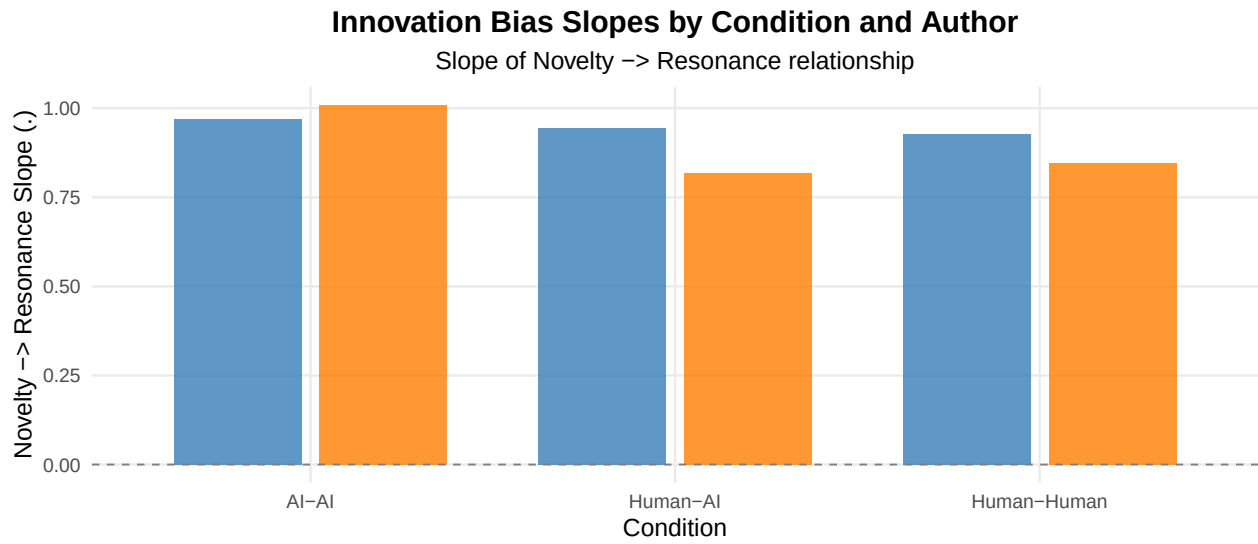
Difference: Author 1 – Author 2. Positive = Author 1 > Author 2



Innovation Bias: Novelty → Resonance Relationship

Steeper slope = novel contributions have greater forward influence





9 Summary

Table 7: Summary: Novelty metric asymmetry (Author 1 - Author 2) by condition

condition	Novelty	Transience	Resonance
Human-AI	0.64 (p=<2e-16) ***	-0.362 (p=<2e-16) ***	1.002 (p=<2e-16) ***
Human-Human	0.045 (p=0.364) ns	-0.025 (p=0.561) ns	0.069 (p=0.264) ns
AI-AI	0.086 (p=0.060) .	-0.05 (p=0.251) ns	0.137 (p=0.036) *

10 Key Findings

Based on the analysis above:

1. **Within-condition asymmetry:** For each condition, we test whether novelty, transience, and resonance differ between Author 1 and Author 2. Significant asymmetry indicates one author contributes more novel/transient/resonant content.
2. **Cross-condition asymmetry comparison:** The $\text{author_role} \times \text{condition}$ interaction tests whether the *pattern* of asymmetry differs across conditions.
 - If Human-AI shows Author 1 > Author 2 novelty but Human-Human shows no difference, the interaction is significant.
 - If all conditions show similar asymmetry patterns, the interaction is non-significant.
3. **Innovation bias asymmetry:** The novelty $\rightarrow$ resonance slope indicates whether novel contributions have lasting forward influence (innovation bias). Testing slope asymmetry reveals whether one author’s innovations are more “rewarded” in narrative development.
4. **Interpretation for Human-AI:**
 - If humans (Author 1) show higher novelty than AI (Author 2), this indicates humans introduce more surprising content.
 - If humans also show higher transience, their contributions fade faster.
 - If humans show higher resonance despite higher transience, their net forward influence is still positive.
 - A steeper innovation slope for humans means their novel contributions disproportionately shape the story.
5. **Baseline comparison:** The AI-AI condition serves as a baseline for symmetric collaboration. Deviations in Human-AI or Human-Human patterns highlight condition-specific asymmetries.

Analysis completed: 2026-04-07 18:49:52.24593